



NON-INTERVENTIONAL STUDY PROTOCOL

STUDY DETAILS

UNIQUE IDENTIFIER	223926
TITLE	Unmet Needs and Rates of Key Background Safety Events of Interest Relating to Treatment Use among Newly Treated and Relapsed Refractory Patients with Multiple Myeloma
STUDY ACCOUNTABLE PERSON	Ashley Holub, Director Epidemiology, Oncology
SCIENTIFIC LEAD	Ashley Holub, Director Epidemiology, Oncology
CONTRIBUTING AUTHORS	Ashley Holub, Director Epidemiology, Oncology Yunji Hwang, Associate Director, Epidemiology, Oncology Kui Huang, Senior Director, Epidemiology Oncology Jessica Petrandis, Director and Lead, Real-World Biostatistics Julia Moore, Associate Director, Real-World Biostatistics Peter Barth, Medical Director, Clinical Oncology Zeyad Khalaf, Medical Director, Global Medical Affairs
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GSK or ViiV ASSET	Belantamab Mafodotin (Blenrep)
EFFECTIVE DATE	26 August 2026
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COMMITMENT

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Title: Ummeet Needs and Rates of Key Background Safety Events of Interest Relating to Treatment Use among Newly Treated and Relapsed/Refractory Patients with Multiple Myeloma

Effective Date: 26 August 2026

Author(s): Ashley Holuh, GSK
Yunji Hwang, GSK

Kui Huang, GSK

Jessica Perhanidis, GSK

Julia Moore, GSK

Peter Barth, GSK

Zeyad Khalaf, GSK

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LIST OF ABBREVIATIONS

Abbreviation	Definition
1L	First-Line Regimen
2L	Second-Line Regimen
3L	Third-Line Regimen
4L	Fourth-Line Regimen
ADC	Antibody-Drug Conjugate
AE	Adverse Event
AST	Aminotransferase
BCMA	B-cell Maturation Antigen
CAR-T	Chimeric Antigen Receptor T-cell Therapy
CCI	Charlson Comorbidity Index
CI	Confidence Interval
CFI	Claims-Based Frailty Index
CRS	Cytokine Release Syndrome
FDA	United States Federal Food and Drug Administration
GSK	GlaxoSmithKline
GU	Genitourinary
Gyn	Gynecological
HIPAA	Health Insurance Portability and Accountability Act
H&N	Head and Neck
ICANS	Immune Effector Cell Associated Neurotoxicity Syndrome
IMiD	Immunomodulatory Agents
LOT	Line of Therapy
NM	Multiple Myeloma
NCCN	National Comprehensive Cancer Network
NDMM	Newly Diagnosed Multiple Myeloma
PI	Proteasome Inhibitors
QC	Quality Control
RRMM	Relapsed Refractory Multiple Myeloma
RWE	Real-World Evidence
RWB	Real-World Biostatistics
SCT	Stem Cell Transplant
SD	Standard Deviation
TE	Transplant Eligible

TI	Transplant Ineligible
TTD	Time to Treatment Discontinuation
TTNT	Time to Next Treatment
ULN	Upper Limit of Normal
US	United States

TRADEMARK INFORMATION

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1. SPONSOR SIGNATORY

Title: Ummet Needs and Rates of Key Background Safety Events of Interest Relating to Treatment Use among Newly Treated and Relapsed Refractory Patients with Multiple Myeloma

Compound Number: GSK2857916

Ashley Holub
Primary Author/NI Scientific Lead
26 Aug 2026
Date (DD Month YYYY)

Kui Huang
Therapy Area Leader/HI Manager
26 Aug 2026
Date (DD Month YYYY)

Note: Not applicable if an eSignature process is used to get the sponsor approval.

2. SYNOPSIS

Title	Ummet Needs and Rates of Key Background Safety Events of Interest Relating to Treatment Use among Newly Treated and Relapsed Refractory Patients with Multiple Myeloma
Background and Rationale	Multiple myeloma (MM) is a progressive clonal plasma-cell malignancy associated with immune dysfunction and recurrent relapse. Within the United States (US) it accounts for roughly 1.7% of newly diagnosed cancers. The five-year survival rate is approximately 64%, with heterogeneous receipt of first-line (1L) and later line treatments often driven by clinical factors such as tolerability concerns. With continued development of the belatnamab (DREAMM-PFI005), and DYNAMIMIC portfolios, RWE is needed to further contextualize effectiveness and tolerability of MM therapies, including background safety rates, and understand unmet need in the treatment landscape.
Study Design and Study Period	This non-interventonal, retrospective cohort study will utilize claims-based data from the Opuma Clinical Data Mart (CDM) database. The study period is defined as 01 Jan 2018 through 31 Mar 2026.
Study Objectives	Primary Objectives 1. To describe the demographics and clinical characteristics of patients with newly diagnosed multiple myeloma (NDMM: 1L) and relapsed/refractory MM (RRMM, 2L and 3L) populations, including background rates of key safety events of interest (i.e., hepatic, infections) and occurrence of healthcare utilization events (i.e., hospitalization and emergency department visits) among patients prior to the receipt of 1L, 2L, and 3L. 2. To estimate incidence of key safety events of interest and healthcare utilization events while on 1L, 2L, and 3L, within each line of treatment (LOT) and by patient subgroups of interest. 3. To describe the occurrence of secondary malignancies following the receipt of therapy. Secondary Objectives 4. To describe treatment patterns and treatment-related outcomes (e.g., attrition, time to next treatment [TTNT] and time to discontinuation [TTD] as proxies for tolerability, overall survival [OS]) as a proxy for effectiveness, by LOT, overall and by SOC, and by patient subgroups of interest. NDMM and RRMM patients who have been treated for MM will be assessed according to the receipt of lines of therapy, according to: Primary analyses: nested cohort • 1L Cohort (i.e., NDMM): newly treated patients in the 1L setting (index = occurring 1L ≥01Jan2019). Receipt of an eligible expected 1L regimen will be utilized as a proxy for NDMM. • 2L Cohort (i.e., RRMM): subset of 1L patients who received 2+ lines of therapy. • 3L Cohort (i.e., RRMM): subset of 2L patients who received 3+ lines of therapy. A secondary 2L (representing 2L+) cohort will also be assessed (2L index ≥ occurring on 01Jan,2020)
Study Population	

1. SPONSOR SIGNATORY

Unmet Needs and Rates of Key Background Safety Events of Interest Relating to Treatment Use among Newly Treated and Relapsed/Refractory Patients with Multiple Myeloma

Title:

Compound Number: GSK2857916

Ashley Holub
Primary Author/NI Scientific Lead
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Kui Huang
Therapy Area Leader/-1 Manager
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Outcomes	Primary objectives: (1) baseline demographic and clinical characteristics, and baseline prevalence of key safety events of interest and healthcare utilization; (2) incidence of key safety events of interest and healthcare utilization while on IL-3L regimens: • Hepatic toxicity • Renal impairment • Severe infection • Cardiovascular events (cardiovascular, thromboembolic events) • Neurological events (movement disorders, neurological conditions) • Other conditions, including secondary malignancies (if applicable) • Healthcare utilization (inpatient admissions, emergency visits) Secondary objective: treatment patterns, and treatment related outcomes to serve as a proxy for efficacy and tolerability: • TTNT • TTD • OS
Statistical Analyses	Study analyses will be descriptive. Descriptive statistics, including mean and standard deviations (SD) for continuous variables, and counts and proportions for categorical variables will be used to summarize baseline data and descriptive outcomes. Prevalence and incidence event rates will be used to estimate occurrences of safety events prior to and while on treatment. Kaplan-Meier curves will be used for time to event outcomes. No formal hypotheses testing will be conducted.

3. AMENDMENTS AND UPDATES

Amendment or update no	Date	Section of study protocol	Amendment or update	Reason
None.	N/A	N/A	N/A	N/A

4. MILESTONES

Milestone	Planned date
Final approval of protocol	31 August 2026
Interim results for ASH 2027	10 January 2027
Final report of study results	01 August 2027

5. BACKGROUND AND RATIONALE

5.1. Background

Multiple myeloma (MM) is a progressive clonal plasma-cell malignancy characterized by immune dysfunction and recurrent relapse, commonly resulting in end-organ damage including lytic bone disease, anemia, renal impairment and hypercalcemia (Morris 2025). MM primarily affects older adults (median age at diagnosis: ~65 years), and remains an important cause of cancer morbidity and mortality worldwide (Rajkumar 2024). In the United States (US), MM represents approximately 1.7% of newly diagnosed cancers, and a similar proportion of cancer deaths (SEER 2026). Although five-year survival has improved over the past two decades, reaching ~64% in recent 2016-2022 estimates (SEER 2026), outcomes remain suboptimal because the disease remains largely incurable and survival varies markedly by age, frailty, and transplant eligibility (Abdallah 2025). Early mortality also remains substantial, particularly among older adults, with approximately 30% of patients dying within one year of diagnosis (Grant 2021).

Against this clinical backdrop, the therapeutic landscape for MM has evolved rapidly, but treatment use, response, and outcomes continue to vary across patients. Frontline care differs by transplant eligibility: transplant-eligible (TE) patients frequently receive triplet induction and autologous stem cell transplant (SCT). Many patients, however, are older or have comorbidities that preclude transplant, though heterogeneity in defining transplant eligibility status remains a non-uniform approach (Neupane 2026). Among transplant-ineligible (TI) patients, frontline regimens vary and are chosen based on patient characteristics and tolerability, with less efficacious outcomes compared to TE patients.

In RRM, CAR-T and bispecific therapies produce deep responses but are limited by access and tolerability. CAR-T is typically offered to younger, fitter patients and is concentrated at specialized academic centres, restricting access by patient fitness and geography (Boysov 2025). Other later line options, such as bispecifics and BCMA-targeted antibody–drug conjugates (ADCs) like belantamab mafodotin, allow broader outpatient access but have distinct safety considerations. BCMA-targeted ADCs are characterised by ocular adverse events, requiring monitoring and dose modifications (Rees 2025). However, ADCs have a lower incidence of T-cell-mediated toxicities, such as cytokine release syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS), as well as parkinsonian neurotoxicity that characterise CAR-T and some bispecifics (Wagner 2025, Wink B 2025).

Compared with more established classes, such as proteasome inhibitors (PI) and immunomodulatory drugs (IMiDs), ADCs have not been characterised primarily by thrombotic or direct cardiotoxic adverse events (Chakraborty R. 2020). However, direct comparative data against bispecific antibodies, PIs, and IMiDs remain limited. Because anti-CD38–based quadruplet regimens often incorporate agents associated with cardiovascular and thrombotic toxicities, comparative evaluation of safety profiles across triplet, quadruplet, and alternative regimens is warranted.

Despite therapeutic advances, unmet needs persist, not only for older and frail patients who have fewer tolerable frontline options, but also in consideration of long-term chronic impacts for younger patients. Trial populations often differ from routine practice, and real-world tolerability and effectiveness can diverge from trial findings. Up-to-date, population-level evidence is therefore needed on background and on-treatment safety rates (for example, ocular, hepatic, and infectious events), healthcare utilization (inpatient and ED visits), treatment

uptake and sequencing, time on therapy, and discontinuation. Administrative claims provide a timely, population-level source for these measures.

5.2. Rationale

Belantamab mafodotin (BLENREP; GSK2857916) is a BCMA-directed ADC evaluated in the DREAMM clinical programme. The pivotal DREAMM-2 trial demonstrated clinically meaningful single-agent responses in a heavily pretreated RRM population, and subsequent DREAMM studies (i.e., DREAMM-7 and DREAMM-8)(ClinicalTrials.gov NCT04246047, ClinicalTrials.gov NCT04484623) have evaluated belantamab-containing combinations and earlier-line use. These trial data illustrate both the potential benefits and the practical challenges of BCMA-targeted ADCs, particularly with respect to tolerability and monitoring (for example, ocular toxicity).

With continued development for DREAMM-10 (ClinicalTrials.gov NCT06679101), PRE1005 (ClinicalTrials.gov NCT07285239), and DYNAMM1C (formerly DREAMM-20) (ClinicalTrials.gov NCT05714839), RWE is needed to further contextualize efficacy and tolerability of alternative non-ADC therapies, including background safety rates, and understand unmet need in the treatment landscape.

This non-interventional, retrospective US real-world study will use the Optum Clinical Datamart (CDM) to estimate incidence of key safety and utilization events before and during therapy, describe treatment patterns and sequencing by line of therapy and SOC, and assess treatment outcomes, including time-to-next treatment (TTNT), time to treatment discontinuation (TTD), and overall survival (OS) as proxies for effectiveness and tolerability among patients with NDMM and RRM. LOT related data will derive from a previously defined LOT algorithm, developed internally by GSK (GSK 2026), summarized in Annex 6.

6. RESEARCH QUESTION AND OBJECTIVE(S)

6.1. Research Question(s)

Investigation of unmet needs with regards to background safety rates according to standard of care (SOC) regimens and treatment utilization and sequencing remains under-characterised. Real-world data can offer insights into safety rates and treatment patterns, which can be used as a proxy for treatment tolerability and efficacy, demonstrating unmet need.

This study aims to address the following research questions, to be addressed according to 3 cohorts of patients across three nested cohorts defined at each LOT: 1L (proxy for NDDMM), 2L (proxy for RRRMM), and 3L (RRMM). Each later LOT cohort is a subset of the prior LOT cohort.

- What are the rates of key safety events (i.e., hepatic events, severe infections, inpatient admissions, secondary malignancies) prior to and during each LOT for the NDDMM and RRRMM settings (i.e., 1L-3L)?
- What are the patterns of treatment receipt, duration, and transition according to standard of care (SOC) regimens/categories in NDDMM and RRRMM patients, to be assessed by LOT?
- How do selected patient characteristics impact treatment use and outcomes?

6.2. Objectives

6.2.1. Primary Objective(s)

The primary objectives of this study will assess unmet needs relating to the MM treatment landscape through assessment of treatment tolerability and efficacy as evaluated through rates of key safety events and treatment patterns in the NDDMM and RRRMM settings. Specifically, the following primary objectives will be addressed:

1. To describe the demographics and clinical characteristics of the NDDMM (1L) and RRRMM (2L and 3L) populations, including background prevalence rates of key safety events of interest (i.e., hepatic, infections) and occurrence of healthcare utilization events among patients prior to the receipt of 1L, 2L, and 3L
2. To estimate incidence of key safety and healthcare utilization events while on 1L, 2L, and 3L, within each LOT and by patient subgroups of interest
3. To describe the occurrence of secondary malignancies following the receipt of therapy

6.2.2. Secondary objective(s)

1. To describe treatment patterns and treatment-related outcomes (e.g., TTNT, TTD, and OS) as a proxy for effectiveness and tolerability by LOT, overall and by SOC, and by patient subgroups of interest

6.2.3. Exploratory objective(s)

1. To assess use of SOC regimens/categories and SCT over time (i.e., SOC type by year)

7. RESEARCH METHODS

7.1. Study Design

This is a real-world retrospective cohort study using the Optum CDM database. The study will describe patient characteristics, safety rates, and treatment patterns, and will utilize treatment related outcomes (TTNT, TTD, OS) as proxies for tolerability and clinical progression/efficacy to inform areas of unmet need. The study period, relevant timepoints, and outcomes for the primary nested cohort are described in Figure 1.

A secondary 2L (RRMM) cohort is later described in **Section 6.4**. All information prior to **Section 7.4** relates to the primary nested cohorts.

- **The study period** will span from 01 Jan 2018 through 31 Mar 2026 (i.e., the most recent date of data availability at time of analysis), and all patients will be required to have initiated their first qualifying line of treatment from 01 Jan 2019.

- The **index date** will be defined as the start date of a **LOT regimen** (i.e., 1L, 2L, 3L; each LOT has its own index date).

- Note: analyses for the primary 3L nested cohort may be limited or excluded pending sample size among SOCs

- The **baseline period** will be defined as the 12-month period prior to the index date for each LOT (does not include index date). Patient demographics, clinical characteristics, and background prevalence safety rates will be summarized during each respective baseline period (i.e., 1L will have a 12-month baseline relative to 1L start, and 2L will have 12-month baseline period relative to 2L start). Only the 1L baseline period will be used to assess study eligibility.

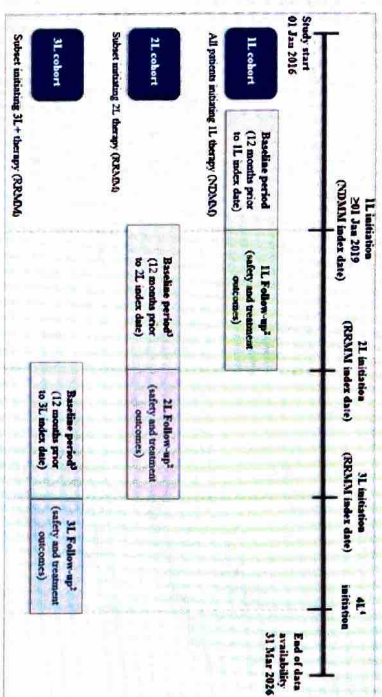
- The patient **follow-up period** will be defined as the period starting from the index date (i.e., including index) until the end of continuous enrollment or end of study period or death, whichever occurs first.

- **Safety and treatment-related outcomes** will be assessed prior to (i.e., during baseline) and during each LOT follow-up period as described in **Section 7.3.2**.

LOT assignment and regimens will be assigned according to prior internal GSK work to define a claims-based LOT regimen (GSK 2026), in brief:

- **1L is defined** as any pre-specified MM therapies received within 60 days of the 1L start date
- **Start of 2L and subsequent LOTs are defined** as the earliest of: a stem cell transplant (SCT; allogeneic or an unplanned autologous SCT), CAR-T cellular therapy, or the date of the first administration for a new MM agent that was not part of the previous LOT regimen. Each subsequent LOT includes all MM therapies received within 30 days on and following the LOT start date

Figure 1. Summary of Primary Nested Cohort Study Design¹



- **Eligible 1L treatment:** Received an eligible or expected treatment for MM on or after MM diagnosis (other than belantamab), occurring on or after 01 Jan 2019 (eligible treatment period).
 - The 1L cohort index date is the date of the first claim for MM treatment within the identification period
 - Eligible/expected treatments include MM regimens commonly used in the first line setting, excluding those restricted to later LOTs
 - Exclusions include: panobinostat and elotuzumab. Other potential therapies pending review of data may be considered
 - For a full list of eligible/expected MM therapies, see **Annex 2**
- **Continuous enrollment (CE):** CE of at least 12-months with medical and pharmacy benefits before the 1L cohort index date. Patients with gaps in enrolment of ≤ 30 days are considered to be continuously enrolled
- **Evidence of follow-up:** at least one claim (pharmacy or medical) from index date or death

Additional eligibility for primary nested 2L and 3L RRM Cohorts

The subset of patients with evidence of each subsequent LOT will be included in the 2L or 3L cohort at the initiation of 2L/3L.

To be eligible for each 2L or 3L cohorts, patients must meet the following criteria:

- **Received a subsequent LOT required to qualify for a specific cohort** (i.e., received a 2L treatment for 2L, 3L for 3L cohort)
- **Continuous enrollment (CE) for each cohort:** CE of at least 12-months with medical and pharmacy benefits before the cohort index date (2L or 3L).

Patients with gaps in enrolment of ≤ 30 days are considered to be continuously enrolled

CE during follow-up for each cohort: at least one claim (pharmacy or medical) from index date

7.2.1.2. Exclusion Criteria

- **Evidence of an MM oncology therapy during the 12-month 1L baseline period:** ≥ 1 medical or pharmacy claim for any MM oncology therapy
 - This is to ensure that treatment exposure and LOT assignments reflect incident therapy starts at the index date and are not confounded by ongoing or recent prior MM treatments
- **Evidence of another cancer in the 1L baseline period:** Patients with either ≥ 1 inpatient or ≥ 2 outpatient ICD-9-CM or ICD-10-CM codes on separate days, within 30 days, for the same primary tumor type and/or metastatic cancer will be excluded
- **Evidence of pregnancy:** ≥ 1 of medical claim with a diagnosis, procedure, or revenue code indicating pregnancy or childbirth during the study period
- **Received belantamab mafodotin (i.e., an ADC) in any LOT**
 - Note: at the time of study belantamab mafodotin was the only ADC in use for MM

7.2.2. SOC Categorization

To assess treatment patterns and time-to-event treatment outcomes according to LOT cohorts and SOC regimen, regimens will potentially be grouped according to commonly utilized quadruplets, triplets, doublets, anti-CD38 backbone, and class, allowing for potential off-label use.

For purposes of inclusion criteria, eligible 1L treatments will include treatments commonly used or expected in the 1L setting, inclusive of approved, and off-label combination regimens.

Annex 2 contains an exemplary list of potential treatment combinations that may be included in each group. Final treatment groupings may depend on data availability and distribution of observed treatments and may be recategorized or summarized as ‘other’, where applicable. Alternative groupings may consider broader categorizations such as “quadruplets, triplets, BCMA vs non-BCMA” or other categories. In addition, if of interest, analyses by selected regimens or treatments may be assessed, pending review of data.

Tentatively, the following SOC categories are proposed:

• **1L (NDMM):**

- Quadruplets with anti-CD38 backbone
- Triplets with anti-CD38 backbone
- Other triplets (non-anti-CD38)
- Doublets/monotherapies
- Other (if applicable)
- **Later lines (2L, 3L, 4L):**
 - Quadruplets with anti-CD38 backbone (off-label; some use expected)
 - Triplets with anti-CD38 backbone
 - Other triplets (non-anti-CD38)
 - Other novel agents (e.g., Selinexor)
 - CAR-T (inclusive of all targets)
 - BCMA bispecific

- Non-BCMA bispecifics
- Doublets/monotherapies (expected to be rare in 2L+)
- Other (if applicable)

7.2.3. Stratification by Patient Subgroups

To assess unmet need according to transplant eligibility, results from the primary and secondary objectives will be stratified by patient subgroups of interest among the 1L and 2L primary nested patient cohorts only. Patient subgroups will be assessed according to LOT and prespecified characteristics. Stratifications with <25 patients will not be performed or may be regrouped due to low volumes.

Planned subgroup stratifications are described below, and are summarized in **Table 1**.

• **By SOC category (described in Section 7.2.2)**

• **Age ≥ 75 vs <75 years**

- Age ≥ 75 years to be defined as proxy for TI status during baseline, to be compared to age <75 years as proxy for TTE

• **Comorbidities of interest**

- Baseline history of neuropathy
- Baseline history of lung parenchymal disease (i.e., COPD, asthma, bronchiectasis, emphysema)
- Baseline history of any event of interest (i.e., cardio, neuro, etc.)

- Baseline frailty status (defined by the Claims-Based Frailty Index [CFI])(Kim 2018)(dependent on data use and mapping availability; Annex 7)
- Potentially other conditions of interest, as determined by review of data

Table 1. Summary of Subgroup Stratifications and Outcomes

No.	Stratification description	Outcomes and cohorts for assessment
1	SOC category	All primary and secondary (SOCs within each LOT)
2	Age ≥ 75 vs <75 years	All primary and secondary by LOT only. 1L and 2L safety and healthcare utilization events at baseline and follow-up only. Age stratification is intended to serve as a proxy for transplant status (where TTE ≥75 years).
3	Neuropathy	Secondary objective by LOT only. 1L and 2L outcomes
4	Frailty status (dependent on data use and mapping availability)	Secondary objective by LOT only. 1L and 2L outcomes

7.2.4. Secondary Malignancy Categorization (Primary Objective 3)

Occurrence of secondary malignancies will be categorized according to clinical relevance. The final groupings will be dependent on review of the data but may consider the following tentative groupings, shown in Table 2.

Table 2. Example Categorization of Secondary Malignancies

Category	Examples
----------	----------

Hematological (will not include other myeloma types)	Leukemia, lymphoma
Genitourinary (GU)	Prostate, renal/kidney, bladder/urothelial, testicular, other GU
Gynecological (Gyn)	Ovarian, endometrial/uterine, cervical, vulva, other gyn
Head and Neck (H&N)	Mucosal squamous cell, nasopharyngeal carcinoma, salivary gland malignancies, other H&N
Gastrointestinal (GI)	Colorectal, gastric/esophageal, pancreatic and hepatobiliary, other GI
Thoracic (Non H&N)	Lung, other thoracic
Breast cancer	Any breast cancer as standalone
Melanoma	Any melanoma
Non-melanoma skin cancer	Basal cell carcinoma, squamous cell carcinoma
Other	Other category dependent on final categorization

7.3. Variables

All variables including treatment, demographic, and clinical variables will be reported descriptively, and will be defined according to available data within Optum CDM.

7.3.1. Exposure Definitions

The exposure of interest is defined as the LOT-based cohort of eligible patients (see Section 7.2.1 for eligibility criteria). Time-to-event treatment outcomes and safety rates during each LOT will be assessed according to SOC regimens and select patient subgroups.

7.3.2. Outcome Definitions

The primary outcomes are key safety events and healthcare utilization during each LOT/cohort baseline and follow-up periods:

- **Primary Objective 1:** summarize demographics, clinical characteristics and background prevalence rates of key safety events of interest during the baseline periods for each 1L, 2L, and 3L cohorts.

- **Primary Objective 2:** assess incidence rates of key safety events while on each LOT (i.e., during the treatment periods)

- **Primary Objective 3:** describe the occurrence of secondary malignancies following the receipt of therapy

Key safety events of interest, as defined in **Table 3**, were selected due to their association with some MM treatments, and will be defined according to selected ICD-10-CM codes or healthcare visits (**Annex 3**).

Table 4 summarizes all outcome definitions, functional form, and timing of collection for the Primary Objectives.

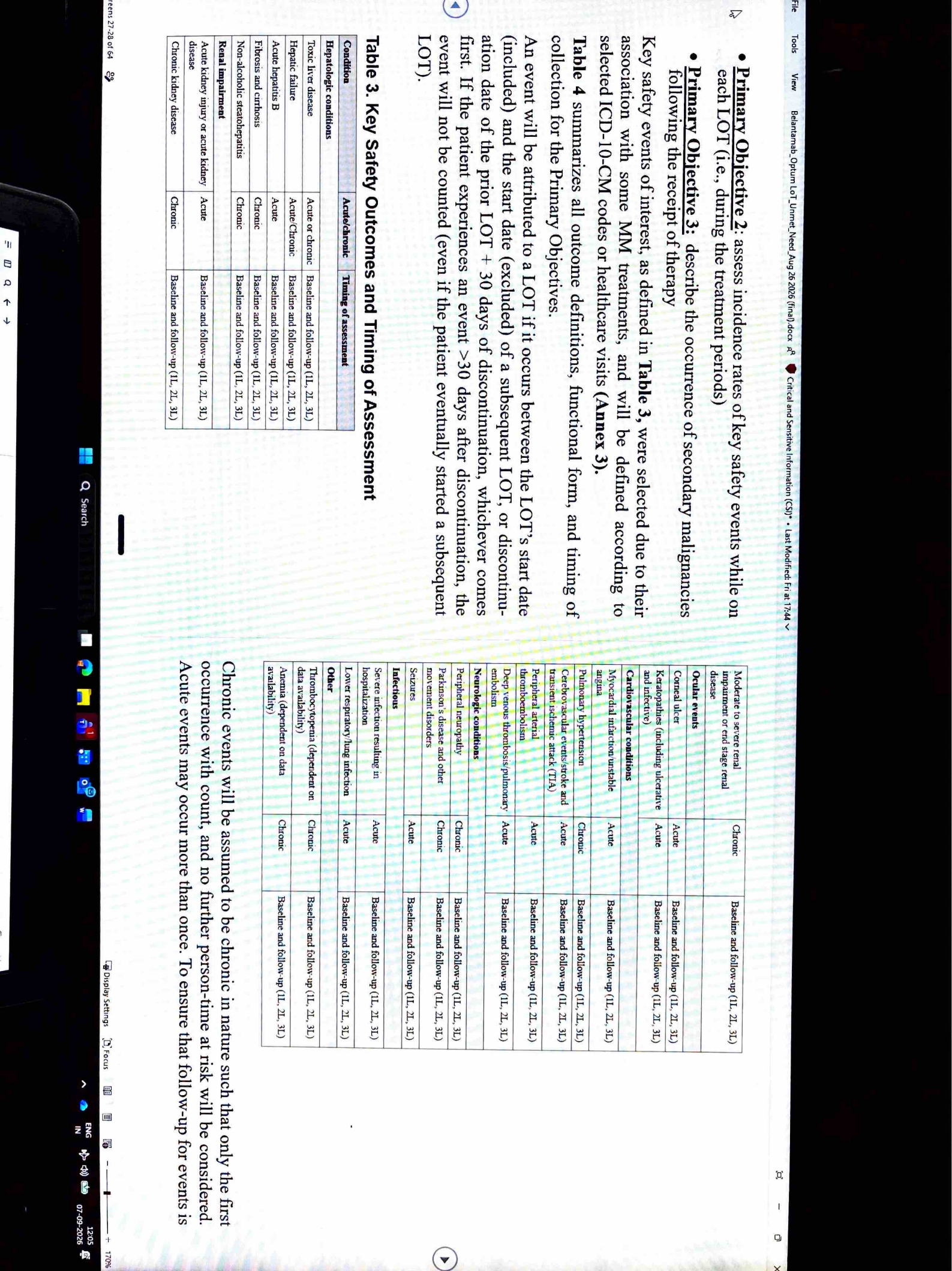
An event will be attributed to a LOT if it occurs between the LOT's start date (included) and the start date (excluded) of a subsequent LOT, or discontinuation date of the prior LOT + 30 days of discontinuation, whichever comes first. If the patient experiences an event >30 days after discontinuation, the event will not be counted (even if the patient eventually started a subsequent LOT).

Table 3. Key Safety Outcomes and Timing of Assessment

Condition	Acute/chronic	Timing of assessment
Hepatologic conditions		
Toxic liver disease	Acute or chronic	Baseline and follow-up (1L, 2L, 3L)
Hepatic failure	Acute/Chronic	Baseline and follow-up (1L, 2L, 3L)
Acute hepatitis B	Acute	Baseline and follow-up (1L, 2L, 3L)
Fibrosis and cirrhosis	Chronic	Baseline and follow-up (1L, 2L, 3L)
Non-alcoholic steatohepatitis	Chronic	Baseline and follow-up (1L, 2L, 3L)
Renal impairment		
Acute kidney injury or acute kidney disease	Acute	Baseline and follow-up (1L, 2L, 3L)
Chronic kidney disease	Chronic	Baseline and follow-up (1L, 2L, 3L)

Moderate to severe renal impairment or end stage renal disease	Chronic	Baseline and follow-up (1L, 2L, 3L)
Ocular events		
Corneal ulcer	Acute	Baseline and follow-up (1L, 2L, 3L)
Keratopathies (including ulcerative and infective)	Acute	Baseline and follow-up (1L, 2L, 3L)
Cardiovascular conditions		
Myocardial infarction/unstable angina	Acute	Baseline and follow-up (1L, 2L, 3L)
Pulmonary hypertension	Chronic	Baseline and follow-up (1L, 2L, 3L)
Cerebrovascular events/stroke and transient ischemic attack (TIA)	Acute	Baseline and follow-up (1L, 2L, 3L)
Peripheral arterial thromboembolism	Acute	Baseline and follow-up (1L, 2L, 3L)
Deep venous thrombosis/pulmonary embolism	Acute	Baseline and follow-up (1L, 2L, 3L)
Neurologic conditions		
Peripheral neuropathy	Chronic	Baseline and follow-up (1L, 2L, 3L)
Parkinson's disease and other movement disorders	Chronic	Baseline and follow-up (1L, 2L, 3L)
Seizures	Acute	Baseline and follow-up (1L, 2L, 3L)
Infections		
Severe infection resulting in hospitalization	Acute	Baseline and follow-up (1L, 2L, 3L)
Lower respiratory/lung infection	Acute	Baseline and follow-up (1L, 2L, 3L)
Other		
Thrombocytopenia (dependent on data availability)	Chronic	Baseline and follow-up (1L, 2L, 3L)
Anemia (dependent on data availability)	Chronic	Baseline and follow-up (1L, 2L, 3L)

Chronic events will be assumed to be chronic in nature such that only the first occurrence with count, and no further person-time at risk will be considered. Acute events may occur more than once. To ensure that follow-up for events is



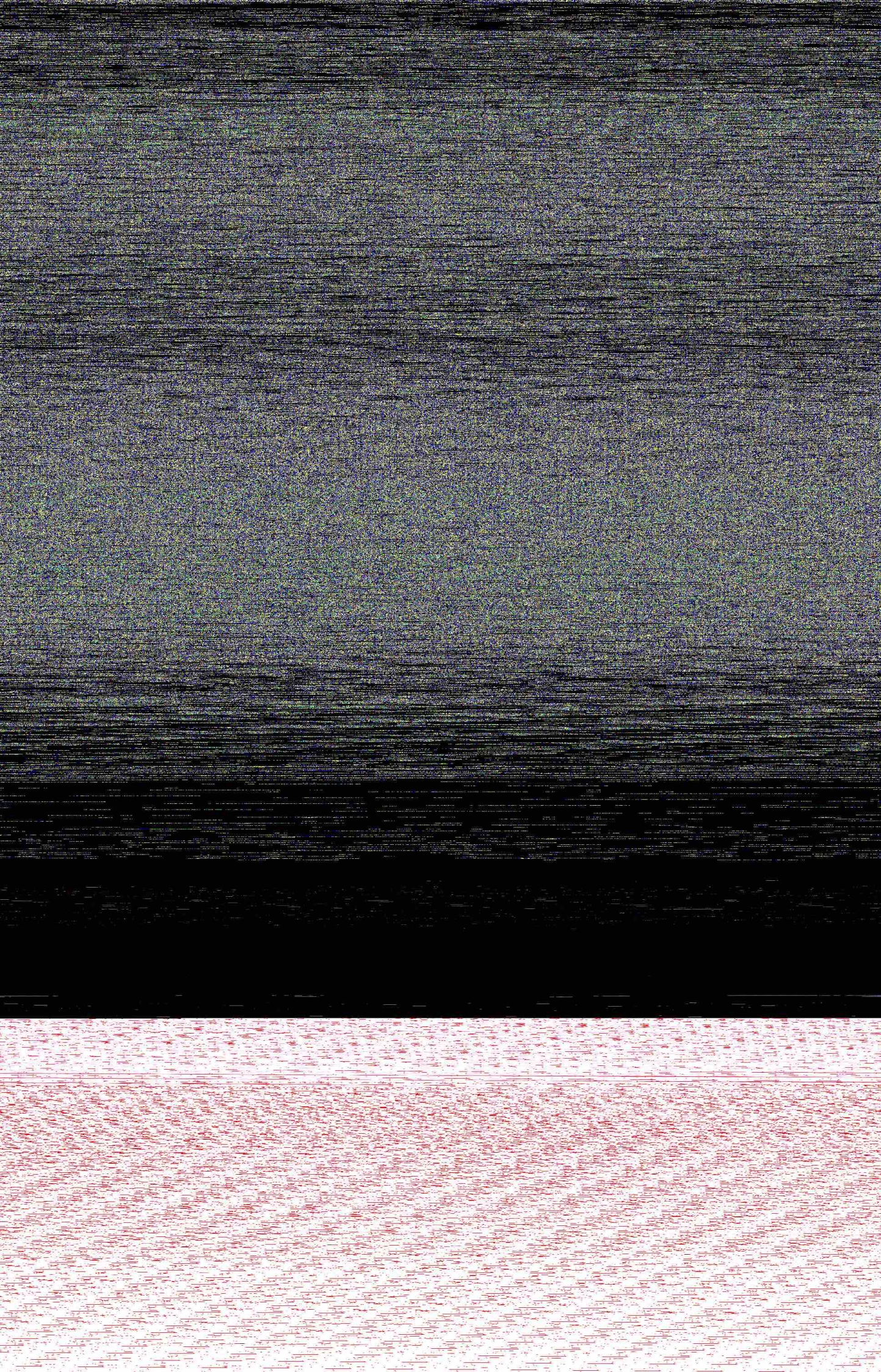
not counted as an event, a ≥ 30 day washout between acute events of the same type is required.

Health care utilization outcomes include: (1) All-cause inpatient hospitalizations and (2) Emergency visits.

Table 4. Baseline Characteristics, Outcomes, Definitions, and Timing of collection for Primary Objectives

Variable/Outcome	Definitions	Timing of Collection
Primary Objective 1: demographics and clinical characteristics of the NDNM (1L) and RRM (2L and 3L) populations, including background prevalence rates of key safety events of interest (i.e., hepatic, infectious) and occurrence of healthcare utilization events among patients prior to the receipt of 1L, 2L, and 3L		
<i>Baseline demographic characteristics</i>		
Age	• Continuous (years) ◦ 18-44 years ◦ 45-64 ◦ 65-74 years ◦ ≥ 75 years <i>Age categories may be adjusted based on age distribution in the study population</i>	At index calendar year (1L, 2L, 3L)
Sex	Categorical: • Male • Female • Unknown	At index (1L, 2L, 3L)
Region	Categorical: • Midwest • South • West • Northeast • Unknown <i>Based on regions defined by US Census Bureau</i>	At index (1L, 2L, 3L)
Race	Categorical: • Asian • Black • White • Unknown	At index (1L, 2L, 3L)

Variable/Outcome	Definitions	Timing of Collection
Ethnicity	Categorical: • Hispanic or Latino • Not Hispanic or Latino • Unknown	At index (1L, 2L, 3L)
Insurance type	Categorical: • Medicare • Commercial Health Plan	At index (1L, 2L, 3L)
<i>Baseline comorbidities and clinical characteristics</i>		
Charlson Comorbidity Index (CCI) (Quan 2011)	Continuous: Categorical: 0, 1, 2, 3, 4, 5+ CCI will be adjusted for having received a MM diagnosis, such that a value of 0 indicates no additional comorbidities beyond MM	During baseline (1L, 2L, 3L)
Kim Frailty Index Score (only included pending review of data and mapping)	Continuous: Categorical: CFI ≥ 0.25 = frail <i>*CFI will only be assessed pending feasibility of inclusion</i>	During baseline (1L, 2L, 3L)
Year of MM diagnosis	Categorical: number and percent of NDNM patients by Year of first MM diagnosis First MM diagnosis is defined as first medical claim for MM within the baseline period on or prior to 1L	First recorded diagnosis date
Follow-up time from diagnosis	Continuous (months): Time from diagnosis date (included) to patient's follow-up end date (included)	Diagnosis to study period
Follow-up time from index	Continuous (months): Time from index date (included) to patient's follow-up end date (included)	Follow-up period (1L, 2L, 3L)
Year of 1L, 2L, and 3L initiation	Categorical: Number and percent of patients by year, from 2019 to latest data availability	At treatment initiation date (1L, 2L, 3L)
Types of 1L, 2L, 3L SOC's or classes by line	Categorical: Number and percent of patients by year, from 2017 to 2025 (or latest data availability)	At treatment initiation date (1L, 2L, 3L)



Variable/Outcome	Definitions	Timing of Collection
Healthcare utilization events	Same as Primary Objective 1	During LOT treatment period (1L, 2L, 3L)
Primary Objective 3: describe the occurrence of secondary malignancies following the receipt of therapy		
Time from diagnosis to secondary malignancy	Continuous (months): Time from diagnosis date (included) until date of secondary malignancy (included)	Diagnosis to occurrence of malignancy
Time from 1L/2L index to secondary malignancy	Continuous (months): Time from 1L or 2L index (included) until date of secondary malignancy (included)	1L and 2L index time to malignancy occurring (2L only among those where the malignancy occurred after 2L)
Prevalence and incidence of secondary malignancy*	To be calculated in the same manner as Objectives 1 and 2, adhering to rules for chronic conditions. *For nested cohort, no background prevalence is needed, for 2L cohort prevalence and incidence are needed	Follow-up period (1L, 2L, 3L)
Type of malignancy	Type of malignancy, defined according to ICD-10-CM codes. Occurrence of malignancy to be confirmed through the presence of at least 2 diagnosis codes occurring on separate dates. The date of the first ICD code will be used	Follow-up period (1L, 2L, 3L)
Secondary malignancy category	Secondary malignancies will be categorized according types, depending on data Pending review of data, the following may be considered: • Hematologic • Head and neck (H&N) • Thoracic (non-H&N) • Genitourinary (GU)	Follow-up period (1L, 2L, 3L)

Variable/Outcome	Definitions	Timing of Collection
	• Gynecological (Gyn) • Gastrointestinal (GI) • Breast Cancer • Melanoma • Non-Melanoma • Other	
LoT after which where malignancy occurred	Defined according to the LoT after which the malignancy is identified	Follow-up period (1L, 2L, 3L)
Top treatment sequences among those with malignancy	Tabulation of the top 5-10 sequences among those with a malignancy occurring after treatment with a malignancy – this will be tabulated among those with a new malignancy occurring only after 2L	Follow-up period (1L, 2L, 3L)

7.3.2.1. Secondary Objective: Treatment Patterns and Treatment-Related Outcomes

The secondary objective to describe treatment patterns and treatment related outcomes will be addressed through descriptive statistics and time-to-event analyses. Thus, unmet need relating to poor treatment outcomes will be proxied through TTNT, TTD, OS, and overall attrition. To further understand unmet need relating to patient characteristics, treatment outcomes will be assessed by patient characteristics (i.e., subgroups).

Table 5 summarizes all outcome definitions, functional form, and timing of collection for the Secondary Objective.

Table 5. Outcomes, Definitions, and Timing of Collection for Secondary Objective

Outcomes	Definitions	Timing of Collection
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Secondary Objective 1: describe treatment patterns and treatment-related outcomes (e.g., treatment attrition, TTNT, TTD, OS) as a proxy for effectiveness and tolerability by LOT, overall and by SOC, and by patient subgroups of interest

Treatment patterns		
Patients receiving each line	Number and percent of patients receiving each 1L, 2L, 3L, and 4L regimens	Follow-up period
Treatment regimens received	Categorical: regimen categories (Section 7.2.2)	Described for overall sequence from 1L to 4L
Switch between successive LOTs	Sankey diagram of switch between regimen categories	Follow-up period
Treatment attrition	Number and percent of patients who received each subsequent LOT, discontinued treatment and did not receive another, were lost to follow-up, or died	Described for overall sequence from 1L up to 4L start
Time from diagnosis to 1L initiation	Continuous (months); Time from diagnosis date (included) until index date (excluded)	1L index
Time from prior LOT to next LOT initiation	Continuous (months); Defined among patients initiating a subsequent LOT as time from prior LOT start date (included) to next LOT start date (excluded)	• 1L start to 2L start • 2L start to 3L start • 3L start to 4L start
Treatment-related outcomes		
Time to next treatment (TTNT)	Time-to-event outcome; Time from index LOT start date (included) to the earliest between the start of the next LOT or death (excluded). Patients without a subsequent LOT or date of death will be censored at their follow-up end date	During each LOT (1L, 2L, 3L)
Time to treatment discontinuation (TTD)	Time-to-event outcome; Time from index LOT start date (included) to the date of treatment discontinuation (excluded). The discontinuation date is the earliest of the date of treatment discontinuation (end of current LOT), initiation of the next LOT, or death. Patients without treatment discontinuation, next LOT or death will be censored at their follow-up end date	During each LOT (1L, 2L, 3L)

**per GSK LOT algorithm definition, discontinuation of a regimen occurs when all MAI agents in the LOT are stopped or when a new agent qualifying SCT event is introduced*

Overall survival (OS)	Time-to-event outcome. Time from LOT start date (included) to date of death (excluded). Patients without a recorded date of death will be censored their follow-up end date	Follow-up period
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7.3.2.2. Exploratory Objectives: Outcomes, Definitions, and Timing
This study will assess several exploratory outcomes, described below in Table 6.

Table 6. Outcomes, Definitions, and Timing of Collection for Exploratory Objectives

Outcomes	Definitions	Timing of Collection
Exploratory Objective 1: assess trends in the use of SCT over time, overall and according to SOC		
Rationale: there is internal need to understand SCT trends and assess prior results		
Trends in the use of SCT	Number of patients with an SCT in 1L–4L by year, according to SOC type	Follow-up period

7.3.3. Confounders and Effect Modifiers
N/A: This analysis is descriptive only.

7.4. Secondary 2L RRMM Cohort

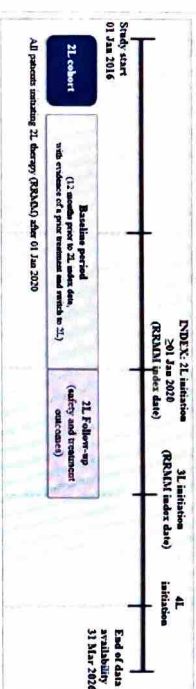
7.4.1. Study Design and Rationale

Due to the rapidly evolving MM treatment landscape, a secondary, non-RRMM 2L + RRMM cohort will be evaluated to assess the robustness of findings from the primary analysis. This cohort will be evaluated using methods consistent with the primary analysis but will include patients with an eligible 2L index

date on or after 01 January 2020, irrespective of whether their 1L initiation occurred during the primary cohort ascertainment period.

The study design is summarized in Figure 2.

Figure 2. Summary of Study Design for Secondary 2L Cohort



7.4.1.1. Inclusion/Exclusion Criteria – Secondary 2L Cohort

A sensitivity analysis in which all patients initiating an assumed 2L therapy on ≥01 Jan 2020 will be assessed.

All inclusion/exclusion criteria will be the same as the primary cohort, with the exception of the index date. Patients in this analysis are eligible if there is evidence of a malignancy prior to 2L.

7.4.1.2. Planned Analyses and Outcomes

All primary and secondary objectives will apply to the secondary cohort. The same outcome definitions as the primary cohort will be applied. All prespecified subgroup analyses will be conducted. No exploratory objectives will be assessed.

The only modification will be to Primary Objective 3 which assesses the occurrence of secondary malignancies. For this objective, all malignancies occurring after diagnosis but prior to 2L will be tabulated, and new malignancies occurring 2L will be assessed.

7.5. Data sources

The Optum CDM database will be utilized for this study. The Optum database is a statistically deidentified, closed system of administrative health claims submitted for payment by providers and pharmacies for US-based commercial health plans and Medicare Advantage members. The database includes the following: (1) patient enrollment; (2) physician, facility, and pharmacy claims. All patients in this database have both medical and pharmacy coverage, allowing analysis of overall healthcare utilization.

Information is available for patients across all care settings including, for example, inpatient visits, emergency department visits, physician office visits (both primary care physician and specialty visits), and prescription medications filled. Available data covers approximately 15-20 million covered lives from 2007 to the present date (estimate current as of 2023) and contains representation from all 50 states. All data is compliant with the Health Insurance Portability and Accountability Act of 1996 (HIPAA). (Optum Insight 2023)

Information for diagnoses of interest, including MM will be collected through claims-based diagnosis tables, while treatment data will derive from pharmacy claims. The pharmacy claims table contains claims submitted by pharmacies for prescriptions filled on an outpatient basis, and includes information on the generic and brand classifications, quantity, and fill dates.

7.6. Study size

Since the main objective of the study is descriptive, it is expected the predicted sample size will be sufficient to produce meaningful results with an appropriate estimate precision. According to a preliminary assessment of available patients who initiated 1L on or after from 01 Jan 2019 through 31 Mar 2026 conducted in August 2026, Optum had approximately 10,514 patients with MM who

received 1L, 5,179 who received 2L, and 3,127 who received 3L. These counts will be further modified and adjusted to meet specific study criteria.

Table 7 presents the levels of precision achieved (i.e., width of estimated 95% CI) with those numbers of patients, for proportion estimates of categorical variables in describing patients' clinical and demographic characteristics of the primary objective. The precision levels were computed using PASS 2019 based on the two-sided 95% Clopper–Pearson exact confidence interval for a single binomial proportion (NCSS Statistical Software 2025).

Table 7. Precision estimates for proportions

Sample size	Proportion (p)	Lower, Upper Limit	Actual 95% Width	Precision (95% CI width, percentage points)
10,514 (1L)	p = 0.1	(0.094, 0.106)	0.012	1.2
5,179 (2L)	p = 0.1	(0.092, 0.108)	0.017	1.6
10,514 (1L)	p = 0.3	(0.291, 0.309)	0.018	1.8
5,179 (2L)	p = 0.3	(0.288, 0.313)	0.025	2.5
10,514 (1L)	p = 0.5	(0.490, 0.510)	0.019	1.9
5,179 (2L)	p = 0.5	(0.486, 0.514)	0.027	2.7

For stratifications (see SOC and patients subgroups stratification in **Sections 7.2.2** and **7.2.3**). Analyses for the primary 3L nested cohort may be limited or excluded pending sample size among SOCs.

7.7. Data management

All data analysis will be conducted using R version 4.5.2.(R Core Team 2025) To derive variables required for analysis, data from Optum CDM will be subset to the population of interest. Information relating to potential eligible patients according to LOT will derive from the previously defined LOT algorithm,

developed in-house by the real-world biostatistics (RWB) team at GSK (GSK 2026).

The GSK RWB team will create the analytical dataset and derive variables from the structured data within Optum CDM required to meet the study objectives. ICD-10-CM codes used to identify eligible diagnoses, and NDC codes to identify treatments will be extracted to define indexes, cohorts, and outcomes. Where applicable, lab values and healthcare utilization data may also be used to defined outcomes.

Data extraction, transformation, and preparation for analysis will be performed using standardized and validated workflows. This includes:

- Data cleaning and validation
- Consistency checks
- Derivation of study variables (e.g., CI, cohort definitions)
- Preparation of analysis dataset
- Quality control

All study data, derived datasets, and analysis outputs will be securely stored and archived in accordance with applicable data protection regulations and internal quality standards. The database is maintained within a secure IT environment, with controlled access, audit trails, and data protection measures in place to ensure confidentiality and integrity.

7.8. Data analysis

Study outputs will be delivered on a sequential, need-driven basis, prioritized according to emerging evidence needs and decision-making priorities. Results not prioritized for formal reporting will remain available for internal review only, via a fit-for-purpose Explorer Tool, and are not intended for external use

unless and until they undergo the process below. Example table shells and figure shells are located in **Annex 4 and Annex 5.**

Any results intended for external use (e.g., regulatory submissions, publications, or other external communications) must be prioritized, undergo appropriate governance review, and receive any additional specification, validation, or quality control needed to support that use, prior to delivery.

All outcomes will be assessed according to their stated objectives. For continuous variables, the descriptive statistics will include means, standard deviations (SD), medians, interquartile ranges (IQR), and minimum and maximum values (as applicable). For categorical variables, frequencies and percentages (%) will be generated. Rates will be described in units of PYs, defined as per 10,000 or 100,000 (or other multiplier), depending on data availability.

The number of patients with unknown/missing values for continuous and categorical variables will be reported. No statistical comparisons or p-values will be reported.

If there are less than 25 patients in a particular stratifications or cohort, analyses will not be conducted (unless specific to SOC).

7.8.1. Primary Objective Analyses

Primary objective 1: demographics and clinical characteristics of the NDDMM (1L) and RMM (2L and 3L) populations, including background prevalence rates of key safety events of interest (i.e., hepatic, infections) and occurrence of healthcare utilization events among patients prior to the receipt of 1L, 2L, and 3L

Patient demographic and clinical characteristics

Baseline demographic and clinical characteristics will be summarized according to the 1L, 2L, and 3L cohorts. These results will be further assessed according to SOC.

All baseline characteristics, except for MM diagnosis date, and comorbidities will be assessed at the time of index date where possible. If data is missing at index, data from the baseline period present nearest index will be used. Comorbidities will be assessed over the 12-month baseline period, including the index date.

Additional subgroup stratifications based on patient characteristics may be applied, depending on review of data and determination of study needs.

Background prevalence rates of key safety events

Background prevalence event rates and corresponding 95% confidence intervals (CIs) will be calculated for each outcome (to be calculated as individual conditions within categories, and aggregated. The total count and percentage of patients experiencing each event will be summarized. Additionally, event rates will be presented for each overall LOT and according to key subgroups of interest.

The numerator will represent the total number of qualifying events for a given outcome. Multiple claims occurring on the same day will be treated as a single event. Events identified on claims occurring more than 1 day apart will be considered distinct events.

The denominator will represent the total amount of PY present in the baseline period (i.e., 12 months prior to each LOT), irrespective of prior event history.

$$\text{Event rate for event type X (per PY)} = \frac{\text{No. of qualifying events X during LOT Y baseline}}{\text{Total baseline PY}}$$

Background rates of health care utilization events

The number and proportion of patients with ≥ 1 hospitalization from any cause, ≥ 1 hospitalization related to MM (defined as having a MM diagnosis in first or second position), or an ER visit during the baseline period will be computed and tabulated for cohorts 1L-3L overall and stratified by SOC.

LOS will be computed from admit date (included) to discharge date (excluded). Hospitalizations that began before the baseline start date or end after the baseline period will be counted at the overall visit level and assigned to the applicable time period. Hospitalizations without a recorded discharge date will be counted when summarizing the number of patients with more than 1 hospitalization and the total number of hospitalizations, but will be excluded from LOS summaries.

In the event of a hospitalization that overlaps baseline/index windows, the assignment will be based on the admit start date. For example, if a patient is admitted 10 days prior to the 1L baseline but is discharged during baseline, the hospitalization will be counted. Likewise, if a patient is admitted during 1L and discharged during 2L treatment, the hospitalization will count only for the 1L treatment period. LOS will be similarly assigned according to the time period in which the admit occurred.

Primary Objective 2: incidence of key safety and healthcare utilization events while on 1L, 2L, and 3L, within each LOT, by categories of SOC regimens, and by patient subgroups of interest

Incidence of safety events

Incidence rates of safety events and 95% CI will be calculated among the overall population in each LOT, SOC, and by key subgroups, according to:

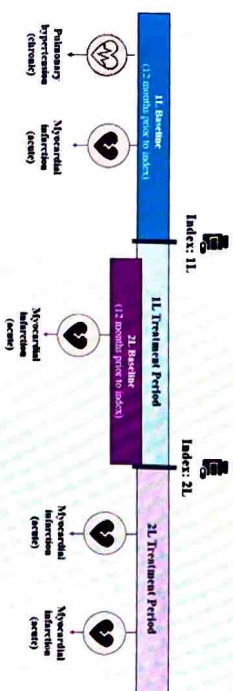
$$\text{Incidence of event type X} = \frac{\text{No. of new event type X occurring LOT Y treatment period}}{\text{Total PY at risk}}$$

Incidence will be calculated as the number of new events occurring during the treatment period. For chronic conditions, only the first occurrence of the condition will be considered an incident event. Individuals with a documented history of the chronic condition prior to the treatment period will not be considered at risk and will be excluded from both the numerator and the person-time denominator for that condition. For acute conditions, each new event occurring during the treatment period will be counted as an incident event.

Chronic events that should only be captured once, at first instance: Chronic kidney disease, Moderate to severe renal impairment or end stage renal disease, Pulmonary hypertension, Peripheral neuropathy, Parkinson's disease, Other movement disorders, malignancies, Thrombocytopenia, Anemia (as defined in Table 3).

An example figure depicting counting of events is shown in Figure 3.

Figure 3. Example Figure of Counting Incident Events



Note: Chronic conditions should only be counted once, at first instance. Acute conditions occurring ≥ 30 days apart may be counted more than once. Hospitalizations due to chronic conditions will be considered an acute event and can be counted more than once.

In this example, the patient has a history of pulmonary hypertension during the IL baseline period. Because pulmonary hypertension is considered a chronic condition, the patient is no longer considered at risk for a new incident pulmonary hypertension event after this baseline occurrence. Therefore, person-time at risk for pulmonary hypertension ends at the time of the baseline event, and no additional pulmonary hypertension events are counted during the remainder of the study.

The patient also experiences a myocardial infarction during the 1L baseline period, which is counted as a baseline event. Because myocardial infarction is considered an acute event, the patient continues to contribute person-time at risk during the 1L treatment period and experiences one myocardial infarction during this period. This event is counted in the 1L treatment period and is also considered part of the patient's medical history at the start of the 2L baseline period.

During the 2L treatment period, the patient continues to contribute person-time at risk for myocardial infarction and experiences two additional myocardial infarctions. These are counted as two separate incident events during the 2L treatment period.

Incidence of healthcare utilization and overlapping hospitalization windows

The number of all-cause hospitalizations, and LOS will be counted according to admit and discharge date. Hospitalizations that occur within the treatment period will be counted.

In the event of a hospitalization that overlaps baseline/index windows, the assignment will be based on the admit start date. For example, if a patient is

admitted 10 days prior to the 1L baseline but is discharged during baseline, the hospitalization will be counted. Likewise, if a patient is admitted during 1L and discharged during 2L treatment, the hospitalization will count only for the 1L treatment period.

Primary objective 3: describe the occurrence of secondary malignancies following the receipt of therapy

The occurrence of secondary malignancies will be evaluated by identifying any new malignancy diagnosed after the initiation of each line of therapy (LoT).

The following metrics and characteristics will be summarized:

- **Time from first observed diagnosis and index to secondary malignancy**
 - Time from first observed MM diagnosis to the onset of the secondary malignancy will be summarized (due to truncation the 2L cohort, this may be assessed in a subset or may not be assessed, pending data assessment)
 - The time from the index date of each LoT (e.g., 1L index, 2L index) to the onset of the secondary malignancy will be calculated and summarized.
- **Incidence (both cohorts) and prevalence (2L cohort only)**
 - **1L nested cohort:** because patients with a baseline history of a prior malignancy are excluded, only the incidence of new secondary malignancies occurring after the initiation of each LoT will be summarized according to type

- **2L cohort:** because prior history of malignancy during baseline is permitted per eligibility criteria, the baseline prevalence of any malignancy will be summarized. Additionally, the incidence of new secondary malignancies occurring after 2L will be summarized by malignancy type
- Refer to **Primary Objectives 1 and 2** for calculation of background prevalence and incidence
- **SOC and treatment sequences**
 - Occurrence of secondary malignancy will be described according to SOC for each LoT
 - The most frequent secondary malignancy types (by category) and the subsequent systemic treatment sequences will be tabulated

7.8.2. Secondary Objectives Analyses

Secondary Objective 1: describe treatment patterns and treatment-related outcomes (e.g., attrition, TTNT, TTD, OS) as a proxy for effectiveness and tolerability by LoT, overall and by SOC, and by patient subgroups of interest

Treatment patterns and regimens received

The proportion of each 1L through 4L regimens received will be described descriptively, as the count and percentage of each regimen.

Treatment sequencing in the overall study sample starting from 1L will be presented using Sankey diagrams, with corresponding tables showing frequency and proportion of treatment type. The Sankey diagram will illustrate transitions between SOC treatments from one LoT to the next, or to no subsequent treatment due to death or censoring at follow-up end date.

Overall patient attrition will be depicted and tabulated. These analyses may be stratified by SOC and patient subgroups.

Time-to-event treatment related outcomes

Outcomes will only be assessed in the subset of patients who have ≥ 3 months of potential follow-up (or die before 3 months) from their index date to ensure adequate time in the database for outcome assessments.

Treatment related time-to-event analyses (e.g., TTNT, TTD, OS) will be performed using the Kaplan-Meier (KM) product limit estimator. Median survival estimates and Brookmeyer-Crowley 95% CI will be the primary estimate reported. (Brookmeyer R 1982) Additionally, the number and percentage of patients at risk, with an event and censored will be reported along with survival probabilities (with 95% CI) at relevant landmarks, such as 6, 9, 12, 18, and 24 months after index date. Time zero, or the index date, will be LoT start/LoT cohort for all TTE outcomes in the primary analyses (i.e., 1L, 2L, 3L). Results will be reported in a tabular format along with corresponding KM curves. No log-rank or hypotheses testing will be performed to assess for differences across strata.

7.8.3. Exploratory analyses

Exploratory analyses are described in **Section 7.3.2.2**. Exploratory analyses will be conducted upon evaluation of data.

7.8.4. Secondary Cohort Analyses (2L Index Cohort)

As noted in **Section 7.4.1.2**, all primary and secondary objectives analyses will apply to the secondary cohort. The same outcome definitions as the primary cohort will be applied. All prespecified subgroup analyses will be conducted. No exploratory objectives will be assessed.

The only modification will be to Primary Objective 3 which assesses the occurrence of secondary malignancies. For this objective, all malignancies occurring after diagnosis but prior to 2L will be tabulated as the background prevalence, and new malignancies occurring 2L will be assessed in line with incidence calculation (See **Primary Objectives 1 and 2 in Section 7.8.1** for calculations).

7.8.5. Data handling conventions

All study outcomes relating to diagnoses will be defined according to pre-defined code lists, as specified in **Annex 3**. Observations where data is missing will be dropped when necessary. No imputation for missing data will be performed.

7.9. Quality control and Quality Assurance

This study will be conducted in accordance with GSK Non-Interventional Studies Standard Operating Procedures.

This study will be conducted according to local quality control processes. This procedure requires documented evidence that the study protocol has been correctly interpreted and executed.

This study will follow established quality control and quality assurance procedures, including the following:

Analytical quality control

- Review of data extraction and transformation processes
- Independent verification of statistical code and outputs
- Reproducibility checks

Data storage and security

- Secure IT environment with controlled access
- Audit trails and data protection measures

- Archiving of datasets and analysis outputs

7.10. Limitations of the research methods

This study should be interpreted in light of its limitations, including limitations inherent to the use of claims-based data. First, this study utilized data available in Optum CDM. Optum primarily offers data from a single private payer, with a small proportion of Medicare advantage plans, thus data is limited to members within these plans and does not represent practices or care receipt among other insurance types or plans. While it is not expected that this restriction would result in different clinical care of patients and receipt of treatments, truncation of data based on payer enrollment may occur. It is possible that patient switching into a plan may have observable data defined as 1L due to this being the first observable treatment, but may have received a prior 1L on another plan before switching in. However, the study inclusion criteria is intended to minimize these instances.

LOTs are not explicitly captured in claims data and the algorithms used to identify lines of therapy were based on pre-defined assumptions. While careful consideration, including alignment to treatment guidelines and expert review was applied to define the algorithm, it is possible that decision rules may not directly align to a patients' medical record or actual care. Regardless, claims-based indicates treatments submitted for reimbursement, which is commonly used as a proxy for receipt of care.

As with retrospective research, the variables and outcomes selected are limited to the underlying data structure and availability, including use of only structured data fields, and absence of confirmatory medical chart data. As a result, TTNT and TTD were used as a proxy for efficacy and tolerability of treatment regimens and further clinical outcomes such as PFS could not be ascertained. As a result, possible explanatory data, such as reasons for discontinuation or

granular adverse event information are not included, but could alternatively explain results. For example, it is possible that a patient who only received IL and stopped had a positive outcome or went into remission, while the current study would interpret this finding as potential poor efficacy. However, due to the nature of MM, it is expected that attrition concerns would likely be negative.

This study has several strengths. Claims-based data offers, large, timely data allowing for increased contemporaneous capture, especially valuable for conditions like MM, where the treatment landscape is rapidly changing. Likewise, claims data is able to capture medication receipt through billing, allowing for ascertainment of treatments. Finally, due to Optum's inclusion of its enrollees, the large sample provided by the database allows for a broader view of the MM population.

8. PROTECTION OF HUMAN SUBJECTS

8.1. Ethical approval and subject consent

This study will comply with all applicable laws regarding participant privacy. No direct subject contact or primary collection of individual human subject data will occur. Study results will be in tabular form and aggregate analyses that omits subject identification, therefore informed consent, ethics committee or IRB approval are not required. Any publications and reports will not include subject identifiers.

8.2. Subject confidentiality

The confidentiality of records that could identify patients within the database must be protected, respecting the privacy and confidentiality rules in accordance with the applicable regulatory requirement(s).

9. LEGAL BASIS FOR PROCESSING INDIVIDUAL HUMAN DATA

The authors confirm that study data is Individual Human Data (IHD) not owned by GSK, but that the proposed use of the IHD aligns with the 'purpose of use' outlined in the source contract and/or the terms and conditions of use of the data source and it will comply with any specified prohibitions of use.

10. MANAGEMENT AND REPORTING OF ADVERSE EVENTS/ADVERSE REACTIONS

Study type 4: Secondary data collection studies based on structured data or unstructured data without human review without safety objectives. Structured data includes de-identified/anonymised healthcare data or health data are converted to a structured forms using automated/computer algorithms e.g. natural language processing.	<i>These studies will not identify safety events as per study objective and cannot identify spontaneous events</i>
Unstructured data without human review, includes use of an automated algorithm and/or natural language processing for data extraction.	

There is no potential to collect serious and non-serious AEs, pregnancy exposures, or incidents related to any GSK product during the conduct of this research, as the minimum criteria of identifiable patient, reporter, exposure and event, needed to collect and report individual case safety reports are not present in the data source. The study is based on secondary anonymised healthcare data which lack an identifiable patient and reporter and are insufficient to establish attribution between a potential safety event and an individual patient using a GSK product.

11. PLANS FOR DISSEMINATING AND COMMUNICATING STUDY RESULTS

Authorship of study manuscripts and presentations at scientific conferences will follow the guidelines established by the International Committee of Medical Journal Editors (<https://www.icmje.org/>). It is intended that one or more peer-reviewed publications will be developed.

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ANNEX 1 LIST OF STAND-ALONE DOCUMENTS

No.	Description	Document Reference No	Date
1.	List of stand-alone documents		
2.	Exemplary categorization of SOC regimens		
3.	Codelists to define study outcomes		
4.	Main study table shells		
5.	Main study figures		
6.	Summary of algorithm to define LoTs		
7.	Claims-based frailty algorithm information and codelists		

ANNEX 2 CATEGORIZATION OF SOC